5. Let's apply this function to one of the PDF files we have to check the content of our document, see *Figure 3.23*. Great! Our function works on one of the ingested documents.

```
with open(f"{documents_folder}2303.10130.pdf", mode="rb") as pdf:
  doc = extract_doc_text(pdf.read())
  print(doc)
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]     /root/nltk_data...
[nltk_data]   Unzipping taggers/averaged_perceptron_tagger.zip.
Sorensen, T., Robinson, J., Rytting, C., Shaw, A., Rogers, K., Delorey, A., Khalil, M., Fulda, N., and Wingate, D. (202
2). An information-theoretic approach to prompt engineering without ground truth labels. In Proceedings of the 60th Annu
al Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). Association for Computational Lingu
istics.
```

Figure 3.23 – Applying a helper function to extract the text from the PDF

Now we need to scale out. We leverage a Pandas **User Defined Function** (**UDF**) to scale the helper function across all pages of all documents.

```
@pandas_udf("array<string>")
def read_as_chunk(batch_iter: Iterator[pd.Series]) -> Iterator[pd.Series]:
    #set llama2 as tokenizer
    set_global_tokenizer(
      AutoTokenizer.from_pretrained("hf-internal-testing/llama-tokenizer")
    )
    #Sentence splitter from llama_index to split on sentences
    splitter = SentenceSplitter(chunk_size=500, chunk_overlap=50)
    def extract_and_split(b):
      txt = extract_doc_text(b)
      nodes = splitter.get_nodes_from_documents([Document(text=txt)])
      return [n.text for n in nodes]

    for x in batch_iter:
        yield x.apply(extract_and_split)
```

Figure 3.24 – Pandas UDF definition for our extractor function

6. Now, let's apply this function to our Delta table, see *Figure 3.25*. We encourage you to expand the content using the > symbol to see the chunk overlap.

Table + New result table: ON Search

| | pdf_name | content |
|---|---|---|
| 1 | dbfs:/Volumes/ml_in_action/rag_chatbot/files/raw_documents/2312.14565.pdf | The Economics of Human Oversight: How Norms and Incentives A |
| 2 | dbfs:/Volumes/ml_in_action/rag_chatbot/files/raw_documents/2312.14565.pdf | However, both groups require more time to complete tasks, with a |
| 3 | dbfs:/Volumes/ml_in_action/rag_chatbot/files/raw_documents/2312.14565.pdf | The high economic value of AI skills (Stephany & Teutloff, 2024) is |
| 4 | dbfs:/Volumes/ml_in_action/rag_chatbot/files/raw_documents/2312.14565.pdf | It contains more than 14 million hand-labelled images of objects ( |

Figure 3.25 – The result from applying our Pandas UDF